Tapeout: setup and submission guide

Nebius × NVIDIA Global AI Hackathon · repo: **github.com/Amritha902/NEBIUS** · deadline: **Oct 30, 10:30 PM IST**

Fastest path: do **Part A** (get 2 keys), then **Part B** (hand them to Claude). While Claude runs the live test, do **Part D step 1** and get ready to record the video.

Part A · Get the two API keys about 10 minutes

1. Nebius Token Factory key (runs the NVIDIA Nemotron models)

1. Go to **https://tokenfactory.nebius.com** and sign up (Google or GitHub login works).
2. If it asks for billing, first open the hackathon's Devpost page and do the to-do item **"Signup for the Nebius Builder Program for credits"**. That gives you free credits.
3. In the console, open **API keys** (left menu or under your profile), then **Create API key**. Name it `tapeout`.
4. **Copy the key immediately.** It is shown only once. Paste it into a private note for now.

2. Tavily key (web search, and the $3,000 Best Use of Tavily prize)

1. Go to **https://tavily.com**, then **Sign up** (free tier is enough).
2. Your key is on the dashboard. It starts with `tvly-`. Copy it.

Never paste keys into the chat, a GitHub file, or a screenshot. They go only into the settings described below.

Part B · Give the keys to Claude about 2 minutes

1. In the Claude session, click the **cloud environment name/menu** in the title bar, then **Edit**.
2. Find **Network secrets** (also called "API credentials" or "Environment variables").
3. Add exactly these two names:

Name	Value
`NEBIUS_API_KEY`	your Nebius key
`TAVILY_API_KEY`	your Tavily key (starts with `tvly-`)

4. Save, then **start a NEW session** in that environment (keys only reach new sessions) and type: **"do the live run for Tapeout in Amritha902/NEBIUS"**

Claude will then:

1. Check that all 3 Nemotron models and Tavily respond.
2. Run the heart-rate example live and fix anything that breaks.
3. Record the live run and rebuild the demo page from it.
4. Push everything and give you the real numbers for the video.

Part C · Run it on your own computer needed to record the video · about 15 minutes

Your computer	Do this first
Windows	Open PowerShell *as administrator*, run `wsl --install`, then restart. Open the **Ubuntu** app and do every step below inside it.
Mac	Install Homebrew from **https://brew.sh** (copy the one-line command from that page into Terminal).
Linux	Nothing extra.

1. Install the tools

Ubuntu / WSL / Linux:

```
sudo apt update && sudo apt install -y git python3 python3-venv ngspice
```

Mac:

```
brew install git python ngspice
```

2. Download the project

```
git clone https://github.com/Amritha902/NEBIUS.git && cd NEBIUS
```

3. Download the chip models (about 2 MB)

```
scripts/fetch_pdk.sh sky130
```

4. Install the app

```
python3 -m venv .venv && .venv/bin/pip install -e '.[dev]'
```

5. Add your keys

```
cp .env.example .env
```

Open `.env` in any text editor. Paste your keys after `NEBIUS_API_KEY=` and `TAVILY_API_KEY=`, then save.

6. Check everything works

```
set -a; . ./.env; set +a; .venv/bin/python -m tapeout.check
```

It must end with **LIVE PATH READY**. If any line says **FAIL**, copy that output to Claude.
No keys yet? `.venv/bin/python -m tapeout.check --offline` checks the simulator alone and should say **SIMULATION PATH READY**.

7. Start the app

```
.venv/bin/uvicorn web.server:app --port 8010
```

Open **http://localhost:8010** and click the **heart-rate (PPG)** example. The banner must say `LIVE NEMOTRON · ACTIVE`. A full run takes 3–5 minutes. **Record your screen during this run** for the video.

Part D · Demo links, video and submission uses your accounts

1. Free instant demo link (GitHub Pages, about 1 minute)

1. Open the repo on GitHub, then **Settings → Pages**.
2. Source: **Deploy from a branch**. Branch: **main**, folder **/docs**. Click **Save**.
3. After about a minute it is live at **https://amritha902.github.io/NEBIUS/demo/**. It plays a recorded run instantly, no server needed.

While you're in Settings: under **Branches**, set the default branch to **main**.

2. Live demo link (optional; Render, about 20 minutes)

1. Sign up at **https://render.com**, then **New → Blueprint**, and pick the **NEBIUS** repo.
2. Add `NEBIUS_API_KEY` and `TAVILY_API_KEY` under **Environment**.
3. The first build takes about 15 minutes. Free instances sleep when idle, so the GitHub Pages link is the reliable one for judges.

3. Record the video (3 minutes or less, with audio)

- Follow the script in `docs/DEMO.md`. Record at 1080p with the browser zoomed to about 110%.
- Say **"NVIDIA Nemotron"** and **"Nebius Token Factory"** out loud at least twice each.
- Must-show shots:
  - the green **LIVE NEMOTRON** banner
  - the red team breaking a claim
  - the **shmoo plot** before vs. after repair
  - clicking a cell to open its real simulation log
  - the **Provenance** table, held on screen for at least 8 seconds
- Upload to **YouTube as Public**, early, because processing can take a while.

4. Submit on Devpost

Field	What to put
Project text	Copy from `docs/DEVPOST.md` in the repo
Track	**Best Apps and Agents**, plus tick **Best Use of Tavily**
Code repository	https://github.com/Amritha902/NEBIUS (public, MIT license)
Demo URL	https://amritha902.github.io/NEBIUS/demo/ (plus the Render link if you made it)
Video	Your public YouTube link
Feedback survey	Fill it in: that's the **Most Valuable Feedback** prize
City (if asked)	Only if you attended a Builders & Brews event

Optional · Show the commits under your name

In your clone (Part C, step 2), run these one at a time:

```
git -c user.name="Amritha S" -c user.email="amritha16112005@gmail.com" rebase --root --exec
'git commit --amend --no-edit --reset-author'
```

```
git push --force-with-lease origin main
```

If Claude has pushed since you cloned, run `git pull` first.

Final checklist

- ☐ Nebius key and Tavily key created
- ☐ Keys added to the Claude environment, new session started, live run done by Claude
- ☐ `tapeout.check` says **LIVE PATH READY** on your computer
- ☐ GitHub Pages on; default branch set to `main`
- ☐ Video recorded (3 min or less) and uploaded to YouTube as Public
- ☐ Devpost: text, track, Tavily prize, repo, demo URL, video, feedback survey, then **Submit**